

UNIT I: HTML5 and CSS

Feature	HTML	HTML5
Doctype	Long and complex	Simplified: <!DOCTYPE html>
Multimedia Support	Requires external plugins (like Flash)	Built-in support for <audio> and <video>
Semantic Elements	Limited semantic tags	Introduced tags like <section>, <article>
Form Enhancements	Basic input types	New types like date, email, range, etc.
Graphics Support	No direct support	<canvas> and <svg> added
Storage	Relies on cookies	Supports localStorage and sessionStorage

A form in HTML is used to collect user input. It sends data to a server using the action and method attributes.

```
<form action="submit.php" method="post">
  <!-- form components -->
</form>
```

Common Form Components:

1. **<input>** – Accepts user input (text, number, checkbox, etc.): `<input type="text" name="username">`
2. **<label>** – Defines a label for inputs: `<label for="username">Username:</label>`
3. **<textarea>** – Multi-line text input: `<textarea name="message"></textarea>`
4. **<select> and <option>** – Dropdown list
`<select name="gender">`
 `<option>Male</option>`
 `<option>Female</option>`
`</select>`
5. **<button> OR <input type="submit">** – Used to submit the form

Common attributes of <input>

1. **type** – Specifies the input type (text, password, email, etc.): `<input type="email">`
2. **name** – Assigns a name for form submission: `<input name="username">`
3. **value** – Defines default value: `<input value="ABC">`
4. **placeholder** – Displays hint text inside input: `<input placeholder="Enter name">`
5. **required** – Makes the field mandatory: `<input required>`
6. **readonly** – Makes input non-editable: `<input readonly value="Admin">`
7. **maxlength** – Sets character limit: `<input maxlength="10">`

limitations of HTML: No Logic/Programming Capability, Static Content Only, Poor Data Handling, Limited Security, Not Suitable for Complex Applications, Browser Dependency

purpose of the <datalist> tag: The <datalist> tag is used to provide a list of predefined options to an <input> element. It helps users by offering auto-complete suggestions while typing.

```
<input list="browsers" name="browser">
<datalist id="browsers">
  <option value="Chrome">
  <option value="Firefox">
  <option value="Edge">
</datalist>
```

Pseudo-Class Selector: A **pseudo-class** is used to define a special state of an element, such as when a user hovers over a link or focuses on an input field.

Common pseudo-classes:

- **:hover** – applies when the user hovers over an element.
- **:focus** – applies when an element (like input) is focused.
- **:first-child** – targets the first child element.

parts of CSS

1. **Selector:** Identifies the HTML element(s) to be styled. Example: p, .class, #id
2. **Declaration Block:** Enclosed in {} and contains one or more **declarations**.
3. **Declaration:** Consists of a **property** and a **value**, separated by a colon. Each declaration ends with a semicolon.